Algorithm for the False-Position Method (Regula Falsi)

Step 1: Input Parameters

- Choose two initial guesses: x0 and x1.
- Set the maximum number of iterations.
- Set the error tolerance.

Step 2: Check if a Root Exists

- If f(x0) ×f(x1) ≥0, there is no guarantee of a root in the given interval. Display a warning and stop execution.

Step 3: Initialize Variables

- Set iteration counter n=1.
- Initialize error er=1

Step 4: Iteration Process

- While n≤ step and er> epes:
    1. Compute the **False-Position formula**:

$$x2=x1-\frac{f(x1)*(x1-x0)}{f(x1)-f(x0)}$$

    2. Compute the relative error: er=|x2−x1/x2|
    3. Update the interval:
        - If f(x0) ×f(x2) <0, set x1=x2.
        - Else, set x0=x2.
    4. Increment n.

Example Walkthrough

Given f(x)=x2−4, with initial guesses x0=1 and x1=3:

f (1) =-3 f (3) =5

f (1) * f (3) <0

find x2

$$x2=x1-\frac{f(x1)*(x1-x0)}{f(x1)-f(x0)}$$

$$x2=3-\frac{5*(3-1)}{5-(-3)}$$

$$x2=3-\frac{10}{8}$$

x2=3- 1.25

x2=1.75

find, f(x2) =1.75^2-4

=-0.9375

Then, check

F(x0) *f(x2) <0

(-) *(-)>0

So,

Else

X0=x2

X0=1.75

Again, find x2